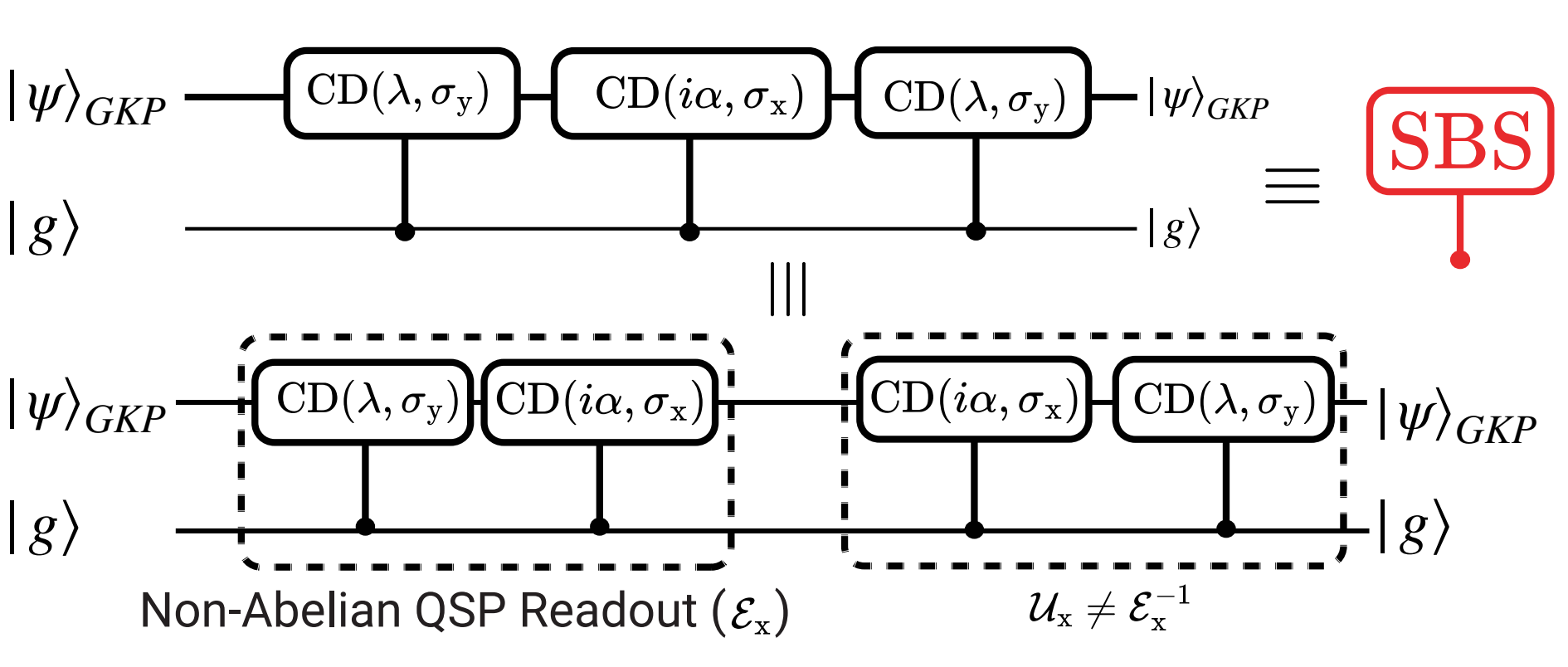

$|\psi\rangle_{GKP}$
$|g\rangle$
$\mathrm{CD}(\lambda, \sigma_y)$
$\mathrm{CD}(i\alpha, \sigma_x)$
$\mathrm{CD}(\lambda, \sigma_y)$
$|\psi\rangle_{GKP}$
$|g\rangle$
≡
SBS
|||
$|\psi\rangle_{GKP}$
$|g\rangle$
$\mathrm{CD}(\lambda, \sigma_y)$
$\mathrm{CD}(i\alpha, \sigma_x)$
$\mathrm{CD}(i\alpha, \sigma_x)$
$\mathrm{CD}(\lambda, \sigma_y)$
$|\psi\rangle_{GKP}$
$|g\rangle$
Non-Abelian QSP Readout ($\mathcal{E}_x$)
$\mathcal{U}_x \neq \mathcal{E}_x^{-1}$